Self-Supervised Learning With Multiscale Densely Connected Network for Hyperspectral Image Classification

Zhen Ye, Zhan Cao, Huan Liu, Haipeng Liu, Wei Li, *Senior Member, IEEE*, Lin Bai, *Member, IEEE*, and Xiaobo Li

***Abstract*— In recent years, deep learning-based methods have exhibited remarkable performance in the field of hyperspectral image (HSI) classification. However, conventional supervised methods heavily rely on a substantial number of labeled samples. Self-supervised learning, as a prominent unsupervised representation learning technique, offers the potential to extract valuable information from unlabeled data. In this article, we introduce a novel unsupervised approach called self-supervised learning with the multiscale densely connected network (SS-MSDCNet) to make full use of unlabeled samples for HSI classification. First, a two-stream structure was designed to generate more positive pairs, which enables the contrast self-supervised training to learn more useful information from unlabeled data. Subsequently, a data augmentation technique based on spectral splitting was proposed to coordinate the two-stream structure of SS-MSDCNet, enhancing spectral information expression. The backbone of the proposed approach is the multiscale densely connected network (MSDCNet), which obtains input HSIs with various spatial scales by removing peripheral pixels from the original input and subsequently extracts multiscale spatial–spectral features using 3-D densely connected modules and 3-D spatial attention modules. The 3-D densely connected module effectively harnesses multiscale features extracted by various convolutional layers, while the 3-D spatial attention module enhances the network's focus on features conducive to accurate classification. To validate the efficacy of our approach, we conducted extensive experiments using four distinct HSI datasets. The results unequivocally demonstrate that SS-MSDCNet outperforms several well-established supervised and unsupervised classification methods. Furthermore, we designed a transfer experiment to confirm SS-MSDCNet's robust generalization capabilities. The code is available at https://github.com/mrblank99/SS-MSDCNet.**

***Index Terms*— DenseNet, hyperspectral image classification, multiscale fusion, self-supervised learning, unsupervised learning.**

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Zhen Ye, Zhan Cao, Haipeng Liu, and Lin Bai are with the School of Electronics and Control Engineering, Chang'an University, Xi'an 710064, China (e-mail: yezhen525@126.com; 2021232008@chd.edu.cn; 2023132025@chd.edu.cn; linbai@chd.edu.cn).

Huan Liu and Wei Li are with the School of Information and Electronics and Beijing Key Laboratory of Fractional Signals and Systems, Beijing Institute of Technology, Beijing 100081, China (e-mail: huanliu233@gmail.com; liwei089@ieee.org).

Xiaobo Li is with China Academy of Space Technology (Xi'an), Xi'an 710100, China (e-mail: lxb619@126.com).

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I. Introduction

HYPERSPECTRAL image (HSI) classification, a critical focus in HSI applications, entails assigning a unique category label to each pixel. Traditional machine learning-based methods for HSI classification predominantly emphasize spectral characteristics and often underutilize the spatial relationships among pixels. Examples of such methods encompass support vector machines (SVMs) [1], sparse representation [2], random forests (RFs) [3], and K-nearest neighbors [4]. Nevertheless, these techniques are plagued by suboptimal classification performance and limited adaptability. With the advent of deep learning, an increasing number of approaches grounded in deep learning are gaining prominence for HSI classification.

In contrast to traditional methodologies, deep learning-based strategies offer superior information extraction and feature learning capabilities [5], [6], [7], [8]. The pioneering work of Chen et al. [9] marked the initial application of convolutional neural networks (CNNs) to HSI classification. This approach involves the pixelwise extraction of fixed-size spatial–spectral blocks to serve as training samples. The exceptional experimental results underscore the importance of spatial information in augmenting the efficacy of HSI classification. To harness the full potential of spatial–spectral information within HSIs, various techniques have been employed, including 3-D CNNs [10], [11], [12], 3-D to 2-D hybrid CNNs [13], and others. These advancements in HSI classification have demonstrated promising results, pushing the boundaries of classification performance and spatial–spectral information utilization.

The previously mentioned deep learning-based approaches have undeniably made significant strides in improving the HSI classification accuracy. However, their remarkable performance often hinges on the availability of a substantial number of labeled samples. In recent years, there has been a growing interest among researchers in self-supervised learning-based methods. In contrast to supervised learning, self-supervised learning has the ability to extract valuable information for classification tasks from unlabeled samples. Several studies have underscored the exceptional classification performance demonstrated in the field of computer vision by self-supervised learning algorithms. With this backdrop, researchers have progressively initiated the application of

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self-supervised methodologies within HSI classification [14], [15]. Self-supervised learning can be categorized into three principal types, contingent upon the nature of the pretraining task: generative, predictive, and contrastive [16]. Generative self-supervised learning [17] operates on the principles of learning data representations through data reconstruction or generation, while predictive self-supervised learning [18] relies on the prediction of self-generated labels to enhance representation learning. Contrastive self-supervised learning [19], [20], [21] focuses on maximizing the similarity between semantically akin inputs as a mechanism for representation learning and obviates the need for pixel-level image reconstruction and the design of intricate auxiliary tasks. Instead, it exclusively involves comparing semantically similar inputs and projecting them closer within the feature space to effectively acquire data representations. Consequently, contrastive self-supervised learning has gained popularity. Well-established contrastive self-supervised learning methods encompass MoCo [22], SimCLR [23], and BYOL [24], among others. A notable development is the work of ContrasNet [25], which integrated contrastive self-supervised learning with autoencoders as an unsupervised feature learning network tailored for HSI classification. Experimental results convincingly underscore the enhanced feature learning capabilities of this network compared to the use of a single autoencoder strategy. Furthermore, Liu et al. [26] have introduced DMVL, which engages in principal component analysis (PCA) on different bands of HSIs to generate multiple views serving as inputs for contrastive self-supervised learning. Subsequently, positive pairs become closer, and negative pairs are farther apart within the projection space. Another notable endeavor is represented by Li et al. [27], who have fused the data augmentation technique of random occlusion with the contrastive self-supervised learning algorithm BYOL to engineer an unsupervised learning algorithm christened ROBYOL for HSI classification. In addition, some novel HSI classification methodologies, such as those combining Transformers and self-supervised learning, have emerged and are gradually gaining recognition [28], [29].

To reduce reliance on annotated samples, this article draws inspiration from SimSiam [30] and introduces a novel two-stream self-supervised learning method for HSI classification. In contrast to existing HSI classification methods that use contrastive self-supervised learning, our approach effectively learns feature representations from unlabeled samples without requiring negative sample pairs, large batch sizes, or momentum encoders. In addition, previous methods for high-dimensional remote sensing image classification based on contrastive self-supervised learning often use models such as ResNet [31] as backbone networks. While these models can efficiently extract features from high-dimensional images, they pass feature maps layer by layer without fully exploiting different-scale features from various convolutional layers. Furthermore, these models typically use fixed-size image blocks as training samples, limiting their ability to fully utilize pixel-level contextual information in high-dimensional images. Therefore, we introduce a multiscale densely connected network (MSDCNet) to extract spatial–spectral features from high-dimensional images by effectively leveraging multiscale features extracted from different convolutional layers through dense connections. Our MSDCNet design provides modules with different image scales by removing peripheral pixels from the input image at each layer, thus fully utilizing pixel-level contextual information. Furthermore, we combine the 3-D densely connected module with the 3-D spatial attention module to extract multiscale spatial–spectral features. Among them, the 3-D densely connected module is obtained by expanding the densely connected module in DenseNet [32] from 2-D space to 3-D space. We refer to this self-supervised learning framework based on MSDCNet as self-supervised learning with MSDCNet (SS-MSDCNet). Moreover, to generate training images for the two streams of SS-MSDCNet, we design a data augmentation technique based on spectral splitting. The designed data augmentation technique is integrated with the two-stream structure of SS-MSDCNet to enhance the network's learning capability. Compared to SimSiam, the proposed SS-MSDCNet utilizes a two-stream architecture and incorporates MSDCNet as its backbone, enabling the extraction of multiscale spatial–spectral features from HSI.

The main contributions of this article can be summarized as follows.

1) We propose a novel two-stream self-supervised learning framework for HSI classification, which enables the backbone network to learn beneficial information from unlabeled data through contrastive self-supervised training. The proposed network completes training without the need for negative sample pairs, which effectively solves the problem of limited computing resources in practical applications.

2) To fully utilize the abundant spectral information of HSIs, a data augmentation technique is designed during the self-supervised training phase. This data augmentation method combines spectral splitting operations with two data augmentation operations, namely, random flipping and random Gaussian noise. Under the support of the two-stream structure, diverse positive sample pairs are generated to learn the similarity of all-band spectral information and the augmented spectral information and then improve the discriminant ability.

3) The network MSDCNet is designed for contrastive self-supervised feature learning. MSDCNet utilizes 3-D densely connected modules and 3-D spatial attention modules to comprehensively explore the spatial–spectral information. Multiscale spatial–spectral features are extracted via a multibranch structure, where the multiscale nature of the feature extraction is finely grained due to feature reuse and spatial attention.

II. Related Work

A. Supervised, Semisupervised, and Few-Shot Learning

Supervised learning entails determining the class of sample pixels through the guidance provided by labeled samples. In recent years, several deep learning-based methods for HSI classification [10], [11], [33], [34], [35] have demonstrated remarkable performance. However, these methods heavily rely on access to a substantial amount of labeled data to achieve exceptional results. Recognizing the challenge of labeled

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data scarcity, researchers have proposed various approaches to HSI classification that integrates elements of semisupervised learning, such as active learning [36], [37], [38], [39]. These approaches utilize both labeled and unlabeled samples during training, leading to enhanced performance by leveraging additional data without extensive labeling efforts. Despite these advancements, the effectiveness of both supervised and semisupervised HSI classification methods remains constrained by the availability of labeled samples. Supervised methods are limited in their ability to learn meaningful representations from unlabeled data, relying solely on labeled samples for training. Semisupervised methods face challenges in designing effective labeling strategies for unlabeled data, often requiring substantial effort in manual annotation or expert knowledge.

To address these limitations, few-shot learning has emerged as a promising alternative [6], [40], [41]. Few-shot learning enables models to generalize from a small number of labeled examples by leveraging prior knowledge or metalearning techniques. This approach mitigates the reliance on large labeled datasets while maintaining robust performance in scenarios where acquiring extensive labeled data is impractical or costly. However, few-shot learning also presents challenges. One major drawback is its sensitivity to the quality and representativeness of the few labeled examples provided. If the labeled examples are not diverse or representative enough, the model's ability to generalize to new classes or tasks may be compromised. In addition, few-shot learning is inherently limited by its inability to utilize large amounts of data, which can restrict its application in tasks where vast datasets are available and beneficial for learning complex patterns or variations.

B. Self-Supervised Learning

In recent years, self-supervised learning has advanced significantly in computer vision. Unlike semisupervised learning and few-shot learning, self-supervised learning effectively leverages extensive amounts of unlabeled HSIs to generate robust feature representations. These representations can be fine-tuned with minimal labeled data, making self-supervised learning particularly suitable for our specific research context. It primarily utilizes pretext tasks to derive intrinsic data representations as supervisory signals and extracts valuable information from abundant unlabeled data. This approach enhances model feature learning capabilities for downstream tasks such as classification, object detection, and semantic segmentation. Self-supervised learning can be categorized into three main types: generative, predictive, and contrastive. Generative self-supervised learning methods focus on learning data representations by either reconstructing or generating input data. This category encompasses techniques such as autoencoders [42], [43], [44] and generative adversarial networks (GANs) [45], [46]. Predictive self-supervised learning methods [47], [48], [49], on the other hand, initiate the process by generating self-defined labels based on designed pretext tasks and then train models to predict these labels as a means of learning data representations. Contrastive self-supervised learning [22], [23] stands out by training models to contrast positive and negative sample pairs, pushing positive pairs closer together in the projection space and driving negative pairs farther apart to optimize data representations. Notably, recent advances in HSI classification have seen the emergence of self-supervised learning-based methods, with a primary reliance on contrastive self-supervised learning. In contrast to generative self-supervised learning, the contrastive approach eliminates the need for pixel-level reconstruction, simplifying network optimization. Furthermore, compared to predictive self-supervised learning, it eradicates the necessity for self-generated labels to create pretext tasks, streamlining the learning process. However, most contrastive self-supervised learning methods applied to HSI classification often require positive and negative sample pairs for training, such as DMVL [26], ContrastiveNet [25], and SC-EADNet [50]. In addition, these methods have not fully used the spectral information of HSI. However, the proposed SS-MSDCNet can complete the training without negative sample pairs, and two-stream structure collaborated data augmentation-based spectral splitting can sufficiently generate positive sample pairs, which is beneficial to spectral feature extraction. Meanwhile, the backbone MSDCNet of SS-MSDCNet, based on 3-D densely connected modules and 3-D spatial attention modules, can effectively utilize the spatial–spectral information of HSI.

III. Methodology

In this section, we provide details of the proposed SS-MSDCNet. SS-MSDCNet commences by engaging in self-supervised training to empower the backbone network with the capability to learn valuable information from unlabeled data that are conducive to classification. Subsequently, the backbone network, refined through self-supervised training, is put to use for feature extraction from HSIs, an essential step in the classification process. To ensure that SS-MSDCNet effectively captures the spatial–spectral characteristics of HSI during self-supervised training, MSDCNet is employed as the foundational network for SS-MSDCNet. Simultaneously, we design a data augmentation method rooted in spectral partitioning, thoughtfully tailored to the specific characteristics of HSIs for application during the self-supervised training phase.

A. Overall Architecture of SS-MSDCNet

The comprehensive framework of the proposed SS-MSDCNet is visually depicted in Fig. 1. This framework comprises two primary phases: two-stream self-supervised training and classification. In the two-stream self-supervised training phase, a substantial volume of unlabeled samples is leveraged to train the backbone network. This process empowers the backbone network to acquire valuable information essential for classification. To make the most of pixel-level contextual information and multiscale spatial–spectral features inherent in HSIs, MSDCNet serves as the backbone network for SS-MSDCNet. It excels in extracting multiscale spatial–spectral features from diverse spatial perspectives. This is achieved by excluding peripheral

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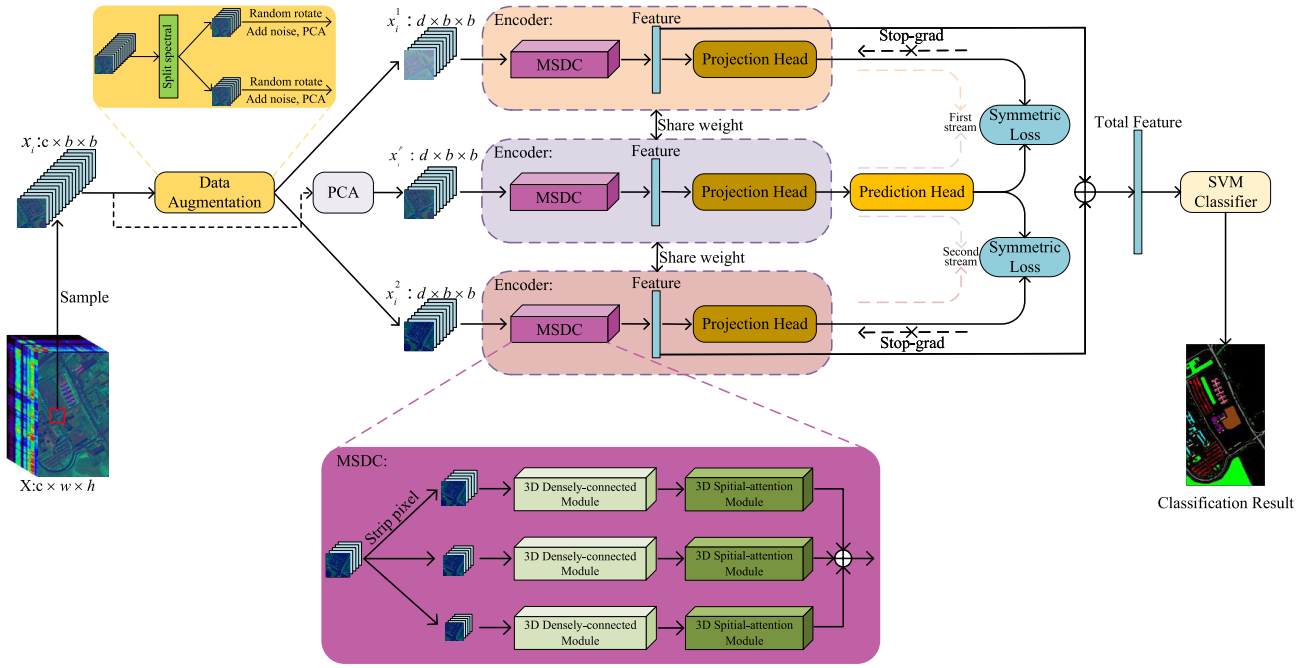


Fig. 1. Overall framework of the SS-MSDCNet including the self-supervised training, MSDCNet as the backbone network, and spectral partitioning as the data augmentation.

pixels using 3-D densely connected modules and 3-D spatial attention modules. During the two-stream self-supervised training phase, it is necessary to perform data augmentation on the splitted HSIs, which correspond to the same ground but exhibit spectral diversity. Thus, we have crafted a data augmentation method centered on spectral segmentation, tailored to the unique attributes of HSIs. In the classification phase, MSDCNet is utilized to extract features from the augmented split HSIs, and the extracted feature maps are then concatenated and used to train an SVM classifier.

B. Self-Supervised Learning of SS-MSDCNet

Inspired by SimSiam [30], we design a two-stream self-supervised network, named SS-MSDCNet. Notably, our approach stands out from other self-supervised methods such as MoCo [22], SimCLR [23], and BYOL [24] due to its lack of dependence on negative sample pairs, large batch sizes, or momentum encoders. Let $\mathbf{X} \in \mathbb{R}^{c\times w\times h}$ denote the HSI, where c, w, and h represent the number of spectral bands, spatial height, and spatial width of $\mathbf{X}$, respectively. To facilitate further processing, we employ a sliding window with the size of $b \times b$ to segment $\mathbf{X}$ pixelwise into patch blocks. Each pixel contributes to the creation of a patch block $\mathbf{x}_i \in \mathbb{R}^{c\times b\times b}$, encompassing both spectral and spatial contextual information. In total, we obtain $w \times h$ patch blocks, where $i \in [1, w \times h]$. Subsequently, these extracted patches are employed for self-supervised training. The designed two-stream self-supervised network involves input data to generate diverse sample pairs. First, we introduce a data augmentation technique rooted in spectral segmentation to create two patch blocks from $\mathbf{x}_i$, denoted as $\mathbf{x}_i^1$ and $\mathbf{x}_i^2$, where $i \in [1, w \times h]$. Next, we process $\mathbf{x}_i$ using PCA to obtain $\mathbf{x}_i^p \in \mathbb{R}^{d\times b\times b}$ (in the proposed SS-MSDCNet, the number of bands b for $\mathbf{x}_i^p$ after PCA is 15). $\mathbf{x}_i^p$ is then combined with $\mathbf{x}_i^1$ and $\mathbf{x}_i^2$ to generate two pairs of positive sample pairs required for self-supervised pretraining.

During the first stream of self-supervised pretraining, the patch blocks $\mathbf{x}_i^1$ and $\mathbf{x}_i^p$ in the first sample pair undergo feature extraction via an encoder network f, which comprises a backbone network and a multilayer perceptron (MLP) projection head. Notably, the encoder network f shares its weights between the two input patch blocks. The output of the encoder network f is defined as

$$\mathbf{z}_i^j = f\left(\mathbf{x}_i^j\right) \tag{1}$$

where $j \in [1, p]$ and the output $\mathbf{z}_i^j$ is processed by an MLP prediction head h to obtain

$$\mathbf{p}_i^j = h\left(f\left(\mathbf{x}_i^j\right)\right). \tag{2}$$

The primary function of MLP prediction head h is to transform the output of one branch to match it with the output of another branch. Since the two streams of self-supervised pretraining have the same structure, the patch blocks $\mathbf{x}_i^2$ and $\mathbf{x}_i^p$ in the second sample pair are also processed by encoder f and prediction head h, respectively.

Early contrastive self-supervised learning makes the encoder learn high-level generic features of images by minimizing the distance between positive sample pairs in the projection space and maximizing the distance between negative sample pairs in the projection space. It typically uses InfoNCE [51] as the loss function to optimize the objective. The computation process of InfoNCE can be described as follows:

$$L_q = -\log\frac{\exp(\mathbf{q}\cdot\mathbf{k}_+/\tau)}{\sum_{i=0}^{k}\exp(\mathbf{q}\cdot\mathbf{k}_i/\tau)} \tag{3}$$

where $\mathbf{q}\cdot\mathbf{k}_+$ represents the cosine similarity between two feature vectors in the positive sample and $\mathbf{q}\cdot\mathbf{k}_i$ represents the

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cosine similarity between two vectors in a negative sample pair. In the two-stream self-supervised training process of the SS-MSDCNet, it does not require negative sample pairs but still manages to prevent model collapse. This is achieved by defining a symmetric loss function different from InfoNCE. Taking the first stream of self-supervised pretraining as an example, the computation process of the symmetric loss function can be described as follows:

$$\hat{L}_1 = \frac{1}{2}D\left(\mathbf{p}_i^1, \mathbf{z}_i^p\right) + \frac{1}{2}D\left(\mathbf{p}_i^p, \mathbf{z}_i^1\right) \tag{4}$$

where D represents the negative cosine similarity, i.e.,

$$D\left(\mathbf{p}_i^1, \mathbf{z}_i^p\right) = -\frac{\mathbf{p}_i^1}{\|\mathbf{p}_i^1\|_2} \cdot \frac{\mathbf{z}_i^p}{\|\mathbf{z}_i^p\|_2} \tag{5}$$

and $\|\cdot\|_2$ denotes the l_2 norm. During the two-stream self-supervised pretraining, another crucial component is the stopgrad during the backpropagation process. Stopgrad can help stabilize the training process and avoid model collapse, even in the absence of negative sample pairs. After adding the stopgrad operation, during the calculation of D, z_i^j will be treated as a constant. The symmetric loss defined above will be rewritten as

$$L_1 = \frac{1}{2}D\left(\mathbf{p}_i^1, \text{stopgrad}\left(\mathbf{z}_i^p\right)\right) + \frac{1}{2}D\left(\mathbf{p}_i^p, \text{stopgrad}\left(\mathbf{z}_i^1\right)\right). \tag{6}$$

Similarly, symmetric loss of the second stream of self-supervised pretraining can be described as

$$L_2 = \frac{1}{2}D\left(\mathbf{p}_i^2, \text{stopgrad}\left(\mathbf{z}_i^p\right)\right) + \frac{1}{2}D\left(\mathbf{p}_i^p, \text{stopgrad}\left(\mathbf{z}_i^2\right)\right). \tag{7}$$

The total loss of self-supervised pretraining is the average of the symmetric losses of two streams. The total loss can be described as

$$L_{\text{total}} = \frac{1}{2}(L_1 + L_2). \tag{8}$$

The pseudocode for the self-supervised training phase of SS-MSDCNet is shown in Algorithm 1.

Algorithm 1 Self-Supervised Training Process of SS-MSDCNet

Input: training dataset $\mathbf{X}$, encoder $f(\cdot)$, prediction MLP head $h(\cdot)$, principal component analysis, i.e., PCA(), data augmentation operation, i.e., agu(), and stopgrad operation, i.e., .detach().

for all $\mathbf{x}_i$ in Dataloader($\mathbf{X}$) **do**
  $\mathbf{x}_i^p = \text{PCA}(\mathbf{x}_i)$
  $\mathbf{x}_i^1, \mathbf{x}_i^2 = \text{agu}(\mathbf{x}_i), \text{agu}(\mathbf{x}_i)$
  $\mathbf{z}_i^p, \mathbf{z}_i^1, \mathbf{z}_i^2 = f(\mathbf{x}_i^p), f(\mathbf{x}_i^1), f(\mathbf{x}_i^2)$
  $\mathbf{p}_i^p, \mathbf{p}_i^1, \mathbf{p}_i^2 = h(\mathbf{z}_i^p), h(\mathbf{z}_i^1), h(\mathbf{z}_i^2)$
  $D(\mathbf{p}_i^1, \mathbf{z}_i^p) = -\frac{\mathbf{p}_i^1}{\|\mathbf{p}_i^1\|_2} \cdot \frac{\mathbf{z}_i^p}{\|\mathbf{z}_i^p\|_2}$
  $D(\mathbf{p}_i^p, \mathbf{z}_i^1) = -\frac{\mathbf{p}_i^p}{\|\mathbf{p}_i^p\|_2} \cdot \frac{\mathbf{z}_i^1}{\|\mathbf{z}_i^1\|_2}$
  $D(\mathbf{p}_i^2, \mathbf{z}_i^p) = -\frac{\mathbf{p}_i^2}{\|\mathbf{p}_i^2\|_2} \cdot \frac{\mathbf{z}_i^p}{\|\mathbf{z}_i^p\|_2}$
  $D(\mathbf{p}_i^p, \mathbf{z}_i^2) = -\frac{\mathbf{p}_i^p}{\|\mathbf{p}_i^p\|_2} \cdot \frac{\mathbf{z}_i^2}{\|\mathbf{z}_i^2\|_2}$
  $L_1 = \frac{1}{2}D(\mathbf{p}_i^1, \mathbf{z}_i^p.\text{detach}()) + \frac{1}{2}D(\mathbf{p}_i^p, \mathbf{z}_i^1.\text{detach}())$
  $L_2 = \frac{1}{2}D(\mathbf{p}_i^2, \mathbf{z}_i^p.\text{detach}()) + \frac{1}{2}D(\mathbf{p}_i^p, \mathbf{z}_i^2.\text{detach}())$
  $L_{total} = \frac{1}{2}(L_1 + L_2)$
  L_{total}.backward()
  update(f.params, h.params)
end for

C. MSDCNet

To facilitate the effective learning of spatial–spectral features within SS-MSDCNet, we introduce MSDCNet as the backbone network that extracts both pixel-level contextual information and multiscale features from HSIs by various convolutional layers.

To capture pixel-level contextual information from HSIs, we obtain different-sized views by removing peripheral pixels from the input image. For instance, in the case of an input patch $\mathbf{x}$ with dimensions of $15 \times 19 \times 19$ (the first dimension is the channel dimension), we derive three distinct views: $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$, resulting in dimensions of $15 \times 19 \times 19$, $15 \times 17 \times 17$, and $15 \times 15 \times 15$, respectively. Given that the input patch is partitioned into three views, MSDCNet takes on a three-branch structure, as illustrated in Fig. 2(a). To ensure consistent feature dimensions for the outputs of these three branches, we initially employ 3-D convolutions with the kernel sizes of $7 \times 11 \times 11$, $7 \times 7 \times 7$, and $7 \times 1 \times 1$ to process $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$, respectively. In these three branches, with the exception of the varying kernel size in the 3-D convolution for feature transformation in the first branch, all other structures remain identical. The feature maps generated by $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ all undergo multiscale feature extraction through a network composed of 3-D densely connected modules and 3-D spatial attention modules.

The 3-D densely connected module proposed in this work comprises seven layers, as depicted in Fig. 2(b). In the first six layers, each includes a $1 \times 1 \times 1.3$-D convolution followed by a $3 \times 3 \times 3.3$-D convolution, while the final layer serves as a feature transformation layer with a $1 \times 1 \times 1.3$-D convolution. The 3-D spatial attention module extends the spatial attention mechanism used in the convolutional block attention module (CBAM) [52] from a 2-D structure to a 3-D structure, as illustrated in Fig. 2(c). CBAM is a lightweight attention module that can perform attention mechanisms on both spatial and channel dimensions, inferring attention weight coefficients along both channel and spatial dimensions. The 3-D spatial attention module initially conducts separate max- and average-pooling operations along the channel dimension on the input feature maps, resulting in two channelwise feature maps, each with a channel dimension of 1. Subsequently, these two feature maps are concatenated, yielding a feature map with a channel dimension of 2. Following this, a 3-D convolution is applied to reduce the channel dimension from 2 to 1. Finally, spatial weight coefficients are generated using the sigmoid function, and these spatial weight coefficients are multiplied with the input feature map to obtain the final output feature map. The 3-D spatial attention module assigns varying weights to the spatial features of each pixel, enabling the network to adaptively extract features more favorable for classification. To convert the feature maps output by the 3-D spatial attention into vectors, a pooling layer and flattening operation are

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TABLE I
DETAILS OF DIFFERENT BRANCH CONVOLUTIONAL LAYERS IN MSDC

	Branch1(Input:1×15×19×19)		Branch2(Input:1×15×17×17)		Branch3(Input:1×15×15×15)	
Layer	Details	Output	Details	Output	Details	Output
Convolution	3Dconv7×11×11	64×8×8×8	3Dconv7×7×7	64×8×8×8	3Dconv7×1×1	64×8×8×8
3D Dense Block	$\begin{bmatrix}\text{3Dconv1}\times1\times1\\ \text{3Dconv3}\times3\times3\end{bmatrix}\times6$	256×8×8×8	$\begin{bmatrix}\text{3Dconv1}\times1\times1\\ \text{3Dconv3}\times3\times3\end{bmatrix}\times6$	256×8×8×8	$\begin{bmatrix}\text{3Dconv1}\times1\times1\\ \text{3Dconv3}\times3\times3\end{bmatrix}\times6$	256×8×8×8
Convolution	3Dconv1×1×1	128×4×4×4	3Dconv1×1×1	128×4×4×4	3Dconv1×1×1	128×4×4×4
3D SA Block	3DSA	128×4×4×4	3DSA	128×4×4×4	3DSA	128×4×4×4
Pooling	3Dmaxpooling4×4×4	128×1×1×1	3Dmaxpooling4×4×4	128×1×1×1	3Dmaxpooling4×4×4	128×1×1×1

employed. These three branch vectors are then concatenated to yield the final output features, as shown in Fig. 2(a).

The output features by the MSDCNet are fed into an MLP projection head for subsequent self-supervised training. Table I summarizes the types and sizes of the three branch convolution layers in MSDCNet, along with the dimensions of the output features from each convolution layer.

D. Data Augmentation Based on Spectral Splitting

In the two-stream self-supervised pretraining stage of SS-MSDCNet, data augmentation is used to generate two distinct samples for $\mathbf{x}_i$. The two generated samples will be combined with $\mathbf{x}_i^p$, respectively, to obtain two positive sample pairs. The choice of appropriate data augmentation techniques is critical in preserving the essential information from the original image. Unlike regular images, HSIs contain rich spectral information, making some data augmentation techniques designed for regular images potentially unsuitable. For instance, techniques such as color jitter may not be suitable. Moreover, HSI classification, being a pixel-level classification task, typically involves inputting image patches that include neighboring pixels into the classification network. The class label of the central pixel in each patch represents the class label for the entire patch. Therefore, data augmentation methods such as random cropping may potentially disrupt the central pixel of the input image, leading to a degradation in classification performance. Given the richness of spectral information in HSIs, this article introduces a data augmentation method based on spectral splitting, as illustrated in Fig. 3. Initially, the HSI is divided into patches along each pixel using a sliding window, as explained in Section III-B, resulting in $w \times h$ patches, each of size $c \times b \times b$, denoted as $\mathbf{x}_i$. Subsequently, a spectral splitting operation is applied along the spectral dimension of $\mathbf{x}_i$, evenly dividing it into two samples with identical spectral dimensions. This approach allows the spectral information from $\mathbf{x}_i$ to be preserved without loss and distributed equally to the two generated samples. Sample augmentation is further achieved by applying random rotations and adding Gaussian noise. Through these two operations, it is possible to increase the dissimilarity between the two samples while still preserving the spectral information. Finally, we process the two samples through PCA, resulting in $\mathbf{x}_i^1, \mathbf{x}_i^2 \in \mathbb{R}^{d\times b\times b}$.

IV. EXPERIMENTAL RESULTS

To evaluate the performance of the proposed SS-MSDCNet, we conduct experiments on four public datasets, including Pavia University, Houston2013, Hyrank-Loukia, and WHU-Hi-HongHu. Initially, we compare the classification performance of SS-MSDCNet with several state-of-the-art HSI classification methods on three datasets, i.e., Pavia University Houston2013 and Hyrank-Loukia datasets. Subsequently, we design transfer experiments based on the WHU-Hi-HongHu dataset. This experiment involves two-stream self-supervised training on the WHU-Hi-HongHu dataset, followed by transferring the pretrained model to the classification of the aforementioned three datasets.

TABLE II
CLASS NAME AND SAMPLE SIZE FOR THE PAVIA UNIVERSITY DATASET

Class No.	Name	Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
	Total	42776

A. Experimental Datasets

1) Pavia University: The Pavia University dataset was collected using the ROSIS sensor and covers the area around the University of Pavia in Northern Italy. It consists of nine classes and has spatial dimensions of 610 × 340 pixels, with a spatial resolution of 1.3 m. The dataset comprises a total of 115 spectral bands, of which 103 bands are considered usable, while the remaining 12 bands are treated as noise and have been removed. Fig. 4 shows the pseudocolor image and the ground-truth map of the Pavia University dataset. The details of the Pavia University dataset are listed in Table II.

2) Houtson2013: The Houston2013 dataset was acquired using the ITRES CASI-1500 sensor. This dataset covers the area around the University of Houston and its vicinity, encompassing fifteen classes. The spatial dimensions of the dataset are 329 × 1905 pixels, with a spatial resolution of 2.5 m. It comprises a total of 144 usable spectral bands. Fig. 5 displays the pseudocolor image and the ground-truth map of the Houston2013 dataset. The details of the Houston2013 dataset are listed in Table III.

3) Hyrank-Loukia: The Hyrank-Loukia dataset was acquired using the EO-1 Hyperion sensor. This dataset was collected as part of an ISPRS scientific project and

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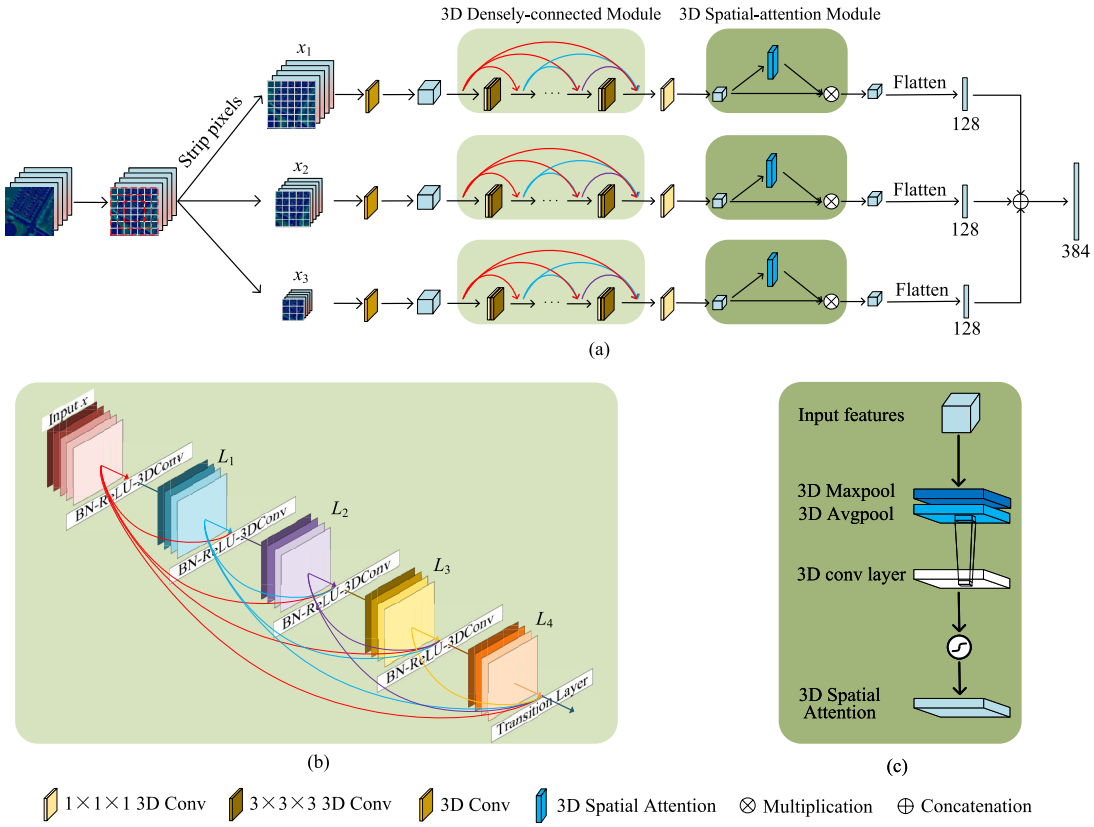


Fig. 2. (a) Structural diagram of the proposed MSDCNet. (b) Details of the 3-D densely connected module. (c) Details of the 3-D spatial-attention module.

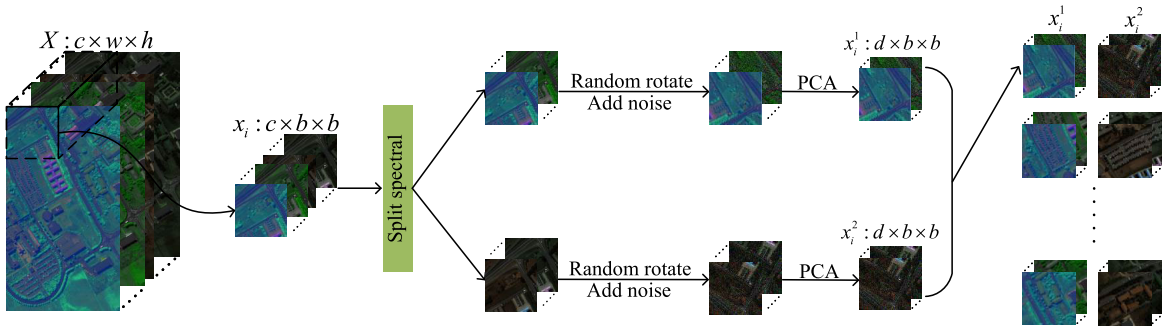


Fig. 3. Specific implementation process of the proposed data augmentation.

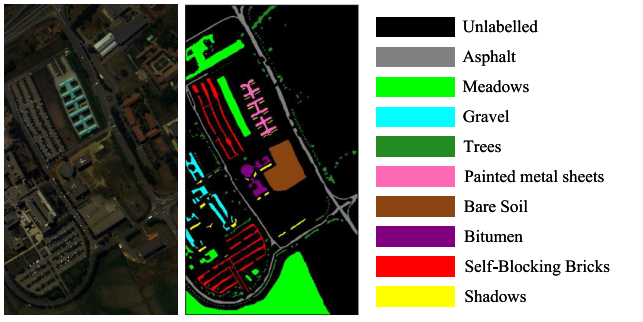


Fig. 4. Pseudocolor image and ground-truth map of the Pavia University dataset.

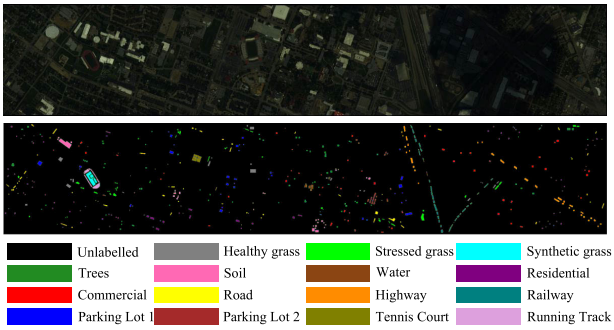


Fig. 5. Pseudocolor image and ground-truth map of the Houston2013 dataset.

consists of 14 classes. The spatial dimensions of the dataset are 250 × 1376 pixels, with a spatial resolution of 30 m. It includes a total of 176 usable spectral bands. Fig. 6 displays the pseudocolor image and the ground-truth map of

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TABLE III
CLASS NAME AND SAMPLE SIZE FOR THE HOUSTON2013 DATASET

Class No.	Name	Samples
1	Healthy grass	1251
2	Stressed grass	1254
3	Synthetic grass	697
4	Trees	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking Lot 1	1233
13	Parking LOt 2	469
14	Tennis Court	428
15	Running Track	660
	Total	15029

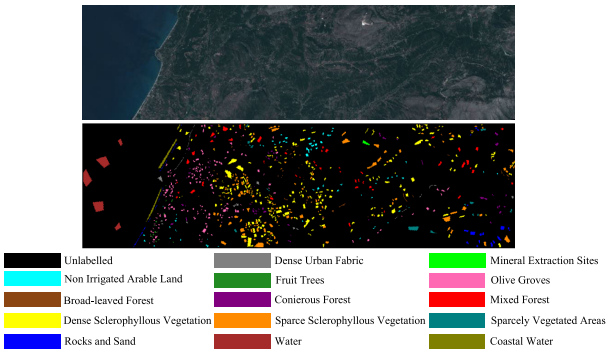


Fig. 6. Pseudocolor image and ground-truth maps of the Hyrank-Loukia dataset.

TABLE IV
CLASS NAME AND SAMPLE SIZE FOR THE HYRANK-LOUKIA DATASET

Class No.	Name	Samples
1	Dense Urban Fabric	288
2	Mineral Extraction Sites	67
3	Non Irrigated Arable Land	542
4	Fruit Trees	79
5	Olive Groves	1401
6	Broad-leaved Forest	223
7	Conierous Forest	500
8	Mixed Forest	1072
9	Dense Sclerophyllous Vegetation	3793
10	Sparce Sclerophyllous Vegetation	2803
11	Sparcely Vegetated Areas	404
12	Rocks and Sand	487
13	Water 2	1393
14	Coastal Water	451
	Total	13503

the Hyrank-Loukia dataset. The details of the Hyrank-Loukia dataset are listed in Table IV.

4) WHU-Hi-HongHu: The WHU-Hi-HongHu dataset was acquired using a 17-mm focal length Headwall Nano-Hyperspec image sensor. This dataset covers a portion of Honghu City in Hubei Province, China, and comprises 22 classes. The spatial dimensions of the dataset are 940 × 475 pixels, with an approximate spatial resolution of 0.043 m. It includes a total of 270 usable spectral bands. Fig. 7 displays the pseudocolor image and the ground-truth map of the WHU-Hi-HongHu dataset. The details of the WHU-Hi-HongHu dataset are listed in Table V.

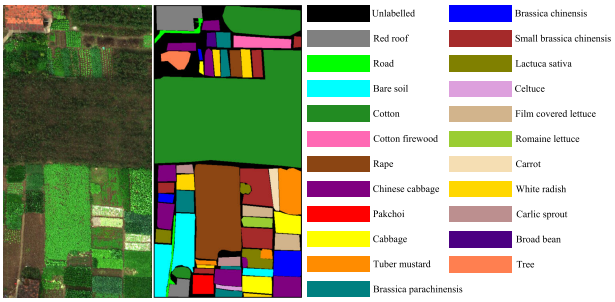


Fig. 7. Pseudocolor image and ground-truth maps of the WHU-Hi-HongHu dataset.

TABLE V
CLASS NAME AND SAMPLE SIZE FOR THE WHU-HI-HONGHU DATASET

Class No.	Name	Samples
1	Red roof	14041
2	Road	3512
3	Bare soil	21821
4	Cotton	163285
5	Cotton firewood	6218
6	Rape	44557
7	Chinese cabbage	24103
8	Pakchoi	4054
9	Cabbage	10819
10	Tuber mustard	12394
11	Brassica parachinensis	11015
12	Brassica chinensis	8954
13	Small Brassica chinensis 2	22507
14	Lactuca sativa	7356
15	Celtuce	1002
16	Film covered lettuce	7262
17	Romaine lettuce	3010
18	Carrot	3217
19	White radish	8712
20	Garlic sprout	3486
21	Broad bean	1328
22	Tree	4040
	Total	386695

B. Experimental Setting

The proposed SS-MSDCNet is implemented using the PyTorch framework. All experiments are conducted on a server equipped with a 3.6-GHz Intel Core i9-9900KF CPU and two NVIDIA GeForce RTX 2080Ti GPUs. The server runs on the Ubuntu 19.04 operating system and has 64 GB of memory. In the experiments, the network utilizes the stochastic gradient descent (SGD) optimizer with a learning rate of 0.03 and is trained for 150 epochs. During the two-stream self-supervised training phase, 50% of the unlabeled samples from the dataset are used as the training set, while the remaining 50% of unlabeled samples are used as the validation set.

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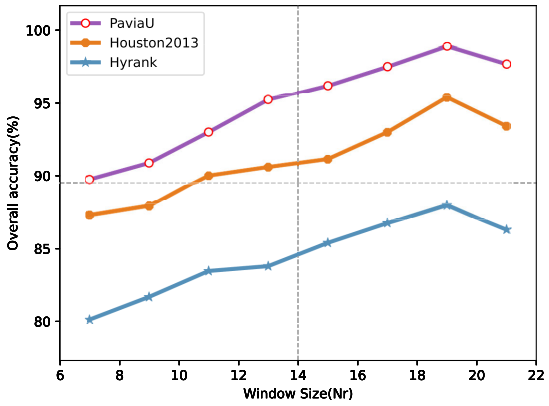


Fig. 8. Classification accuracies for different window sizes.

During the classification phase, 5% of labeled samples are randomly selected from each class to train the supervised classifier for the Pavia University and Houston2013 datasets and 10% for the Hyrank-Loukia dataset. Overall accuracy (OA), average accuracy (AA), and Kappa are utilized to measure the classification performance. To mitigate potential biases arising from sampling, all performance metrics are computed as the mean and standard deviation over ten runs.

As shown in Fig. 1, HSI is divided into patches using a sliding window of size $b \times b$. To analyze the impact of window size on classification accuracy, we experiment with window sizes in set $\{i \times i; i = 7 : 2 : 21\}$. As illustrated in Fig. 8, SS-MSDCNet achieves the best performance when the window size is set to 19 for the Pavia University, Houston2013, and Hyrank-Loukia datasets. When the window size is increased to 21, the OA for all datasets decreases. This might be attributed to the patch size being too large, resulting in patches containing samples from different classes. Therefore, the window size is fixed as 19 in the following experiments.

C. Comparison With Different Methods

We perform two-stream self-supervised pretraining on three public datasets separately to validate the classification performance of the proposed SS-MSDCNet. Then, we compare the performance of the proposed SS-MSDCNet with several state-of-the-art methods. These methods include supervised, semisupervised, unsupervised, and self-supervised methods. In addition, we also train the designed backbone network MSDCNet in a supervised manner on all three datasets for comparison. The compared methods are listed as follows.

1) SVM [1] is a classical supervised method for HSI classification. The method uses the RBF kernel function. The optimal values of hyperparameters C and γ are determined by fivefold cross validation in the range of 2^{-2}–2^{9}.
2) 3-D CNN [10] is a supervised method to extract joint spatial–spectral features of HSI by 3-D CNN. The window size is set as 3×3.
3) BASSNet [53] is a supervised method for extracting band-specific spatial–spectral features of HSI to accomplish classification of HSI. The window size is set as 5×5.
4) MSDCNet is a supervised method that trains the designed backbone network in SS-MSDCNet in a supervised manner. The method reduces the spectral dimension to 15 by PCA, and the window size is set to 19×19.
5) ALPN [54] is a semisupervised HSI classification method that combines semisupervised clustering and active learning. The method reduces the spectral dimension to 15 by PCA, and the window size is set to 15×15.
6) Semi-LG-AGCN [38] is a semisupervised network with end-to-end local-global active learning based on graph convolutional network. The window size is set as 7×7.
7) DSCA-Net [55] is a semisupervised HSI classification method based on the two-stream class adaptive network. The method can suppress the negative effects of uneven data distribution by a superpixel-guided label propagation module. The window size is set to 24×24.
8) VAE [43] is an unsupervised method that introduces probabilistic graphical models and random variables into autoencoder so that VAE can learn the distribution of high-dimensional data. The method reduces the spectral dimension by PCA to 15, and the window size is set to 15×15.
9) SimSiam [30] is a contrastive self-supervised method. The method learns the potential features of the data from unlabeled data through the Siamese network. The method also reduces the spectral dimension to 15 by PCA, and the window size is set to 19×19.
10) DMVL [26] is a method for HSI classification based on contrastive self-supervised learning. The method enables deep networks to learn view-invariant features. The method reduces the spectral dimension to 3 by PCA, and the window size is set to 27×27.
11) ContrastiveNet [25] is an HSI classification method that combines contrastive self-supervised learning and autoencoder. The method uses features extracted from HSI by two different autoencoders, VAE and AAE, as input to the Siamese network. The method also reduces the spectral dimension to 15 by PCA, and the window size is set to 15×15.
12) SC-EADNet [50] is a contrastive self-supervised learning method using an efficient asymmetric dilated network as the backbone. The method reduces the spectral dimension to 3 by PCA, and the window size is set to 31×31.

For the Pavia University dataset and the Houston2013 dataset, the supervised or semisupervised comparative methods use 5% of the labeled data as the training set and the remaining 95% of the labeled data as the test set. For the Hyrank-Loukia dataset, the supervised or semisupervised comparative methods use 10% of the labeled data as the training set and the remaining 90% of the labeled data as the test set. For all datasets, unsupervised or self-supervised comparison methods use the same dataset partitioning strategy as SS-MSDCNet.

In Tables VI–VIII, among the supervised or semisupervised methods, the proposed backbone, i.e. MSDCNet, performs better than most comparison baselines on all three datasets. This confirms its superior performance in

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TABLE VI

CLASSIFICATION RESULTS (%) FOR THE PAVIA UNIVERSITY DATASET (5% LABELED SAMPLE PER CLASS)

Class No.	Supervised methods or semi-supervised methods							Unsupervised or self-supervised methods					
	SVM [1]	3D-CNN [10]	BASSNet [53]	MSDCNet	ALPN [54]	Semi-LG-AGCN [38]	DSCA-Net [55]	VAE [43]	SimSiam [30]	DMVL [26]	ContrastiveNet [25]	SC-EADNet [50]	SS-MSDCNet (SVM)
1	83.54±0.28	97.51±1.14	94.67±1.27	96.98±2.54	87.32±5.22	97.61±1.63	**99.26±0.67**	77.73±1.06	95.21±0.43	93.96±0.55	92.49±1.02	93.89±2.06	97.75±0.38
2	88.26±0.15	96.33±0.85	99.61±0.20	**100.00±0.00**	97.58±2.32	99.75±0.27	99.84±0.17	97.50±0.83	94.06±0.45	98.46±0.31	97.05±0.68	92.24±1.80	99.72±0.15
3	54.31±3.26	94.12±3.26	76.21±5.58	96.63±1.61	88.39±4.79	79.38±11.97	**98.77±0.84**	87.61±2.05	96.32±0.64	94.94±1.24	92.36±1.94	88.07±5.57	96.37±1.54
4	86.57±1.40	97.36±2.17	95.99±0.98	87.60 ±5.49	93.41±5.31	94.92±4.38	93.44±1.24	92.78±2.15	94.75±1.43	87.18±2.79	88.22±1.48	**98.74±1.79**	97.63±0.80
5	99.03±0.22	98.24±0.37	99.52±0.48	**99.98±0.07**	99.80±0.40	99.84±0.29	99.65±0.47	99.86±0.18	98.97±0.42	86.32±0.14	97.21±0.97	99.87±0.22	99.48±0.30
6	61.15±1.03	98.31±1.74	95.35±1.24	92.25±11.05	96.25±3.04	97.38±1.49	99.98±0.06	**99.99±0.03**	98.08±0.57	97.43±0.70	97.25±1.02	86.76±6.83	99.82±0.07
7	56.14±0.27	97.02±3.16	81.81±5.13	97.90±2.88	98.82±1.17	98.60±2.18	98.81±2.11	98.68±0.73	97.61±1.36	93.34±1.47	92.78±1.67	94.47±4.10	**99.54±0.77**
8	75.28±0.54	97.32±2.81	92.74±3.06	95.53±3.69	82.62±13.29	96.05±3.45	97.67±0.84	92.63±0.97	95.81±0.67	**97.87±0.51**	93.70±1.18	93.61±3.15	97.54±1.02
9	99.91±0.11	**99.99±0.34**	99.41±0.43	94.67±3.02	97.08±2.34	98.18±1.98	86.28±5.16	87.23±1.39	93.35±2.73	85.12±3.32	89.66±3.09	99.96±0.11	98.00±1.27
OA(%)	78.92±0.11	95.44±1.05	95.78±0.52	97.00±1.44	93.90±1.37	97.41±0.70	98.73±0.19	93.37±0.42	95.23±0.18	95.85±0.34	94.92±0.35	92.57±0.32	**98.87±0.16**
AA(%)	76.16±0.58	94.01±0.72	92.81±0.88	95.73±1.53	93.47±1.66	95.76±1.28	97.08±0.67	92.67±0.31	96.02±0.35	92.74±0.60	93.41±0.44	94.18±0.72	**98.43±0.33**
kappa(%)	70.52±0.20	94.03±1.42	94.40±0.69	95.98±1.97	91.91±1.82	96.56±0.92	98.32±0.25	91.28±0.54	93.61±0.25	94.49±0.45	93.28±0.46	90.05±0.46	**98.51±0.21**

TABLE VII

CLASSIFICATION RESULTS (%) FOR THE HOUSTON2013 DATASET (5% LABELED SAMPLE PER CLASS)

Class No.	Supervised methods or semi-supervised methods							Unsupervised or self-supervised methods					
	SVM [1]	3D-CNN [10]	BASSNet [53]	MSDCNet	ALPN [54]	Semi-LG-AGCN [38]	DSCA-Net [55]	VAE [43]	SimSiam [30]	DMVL [26]	ContrastiveNet [25]	SC-EADNet [50]	SS-MSDCNet (SVM)
1	90.21±0.58	94.72±1.34	**98.38±0.64**	93.73±2.56	86.03±7.59	97.85±2.94	92.81±3.87	88.20±4.11	86.49±3.80	88.83±3.75	93.13±3.26	90.36±3.22	97.45±1.31
2	91.26±0.94	98.26±0.57	**99.02±0.77**	98.63±0.69	95.23±4.29	96.55±4.06	94.72±2.96	88.55±3.63	91.36±2.22	82.55±4.17	96.91±3.09	92.44±3.73	95.38±2.04
3	99.03±0.35	99.21±0.45	99.77±0.17	93.43±1.42	99.04±0.95	**99.96±0.07**	97.35±2.19	99.68±0.40	99.14±0.74	97.95±1.75	98.66±1.23	98.00±4.05	97.03±0.04
4	95.83±0.39	**98.39±0.53**	93.59±1.78	92.70±1.60	93.75±2.62	95.92±2.29	92.88±4.70	87.61±3.33	87.15±2.80	80.29±2.43	88.90±3.82	94.92±3.26	95.95±0.71
5	94.26±0.51	98.41±0.95	99.07±0.65	**100.00±0.00**	98.93±1.95	99.99±0.03	99.82±0.35	99.69±0.25	95.96±1.01	93.98±2.14	96.56±2.10	98.05±0.99	96.19±1.08
6	90.58±0.98	95.41±1.24	67.22±4.82	82.49±1.17	92.76±5.41	94.46±6.34	83.23±5.45	88.19±5.91	95.08±3.30	83.11±5.99	77.54±5.41	**96.35±4.65**	95.84±2.23
7	73.38±0.62	92.31±0.94	89.25±2.23	81.66±6.33	77.01±5.67	91.34±3.63	**93.80±3.25**	83.24±3.70	88.07±3.25	81.19±2.55	84.01±4.64	89.23±4.43	92.69±1.37
8	64.66±1.61	83.33±2.71	78.52±2.03	91.07±1.90	71.98±7.20	73.03±9.38	86.67±6.41	81.67±3.44	87.71±3.70	90.35±2.93	88.16±2.23	80.13±3.62	**93.86±2.48**
9	68.73±0.85	87.08±1.99	80.24±2.46	88.85±3.65	80.39±7.27	87.91±3.94	92.58±2.60	79.76±2.23	87.40±2.36	87.15±2.76	85.61±4.48	83.13±2.05	**93.60±1.94**
10	55.69±2.05	89.71±1.76	91.37±2.71	99.31±0.95	93.84±4.88	96.39±4.43	**99.78±0.41**	87.42±7.52	89.73±2.78	97.51±1.92	91.75±3.84	85.57±2.27	96.41±1.13
11	60.72±1.73	88.76±4.39	84.72±4.56	93.69±3.00	94.56±4.32	91.64±4.40	95.91±3.47	87.41±3.17	91.98±2.27	**97.46±1.11**	96.05±1.34	90.86±2.88	94.61±2.12
12	55.96±1.38	90.26±3.76	91.12±2.13	**99.56±0.36**	91.21±2.82	88.82±6.31	95.98±3.19	89.51±4.19	92.08±3.40	91.68±3.08	91.84±3.33	87.20±4.54	94.91±1.97
13	51.80±2.79	77.52±5.83	47.42±9.50	94.04±3.77	95.98±2.11	85.67±2.78	93.41±3.78	85.31±6.19	95.81±1.75	89.20±5.53	84.98±7.10	91.63±1.37	**96.12±1.37**
14	86.27±1.08	98.48±1.21	91.08±3.04	**100.00±0.00**	99.79±0.26	99.95±0.10	**100.00±0.00**	99.02±1.04	99.01±0.86	99.88±0.12	95.16±3.92	91.62±3.17	99.90±0.15
15	98.82±0.51	99.16±0.32	95.71±2.41	**100.00±0.00**	98.51±4.42	**100.00±0.00**	97.37±1.68	99.84±0.43	95.86±2.73	96.41±3.13	98.25±2.37	97.04±4.66	98.05±1.49
OA(%)	75.78±0.31	92.45±0.64	89.35±0.55	94.06±0.74	89.83±0.92	92.76±0.91	94.62±0.91	88.71±0.90	90.91±0.60	89.87±1.23	91.54±0.55	89.26±1.00	**95.47±0.13**
AA(%)	75.33±0.69	91.91±0.35	87.10±0.79	93.94±0.65	91.27±0.70	93.30±0.78	94.42±0.78	89.67±0.84	92.19±0.57	90.50±1.36	91.17±0.72	91.10±1.28	**95.87±0.12**
kappa(%)	73.66±0.32	91.46±0.38	88.47±0.60	93.58±0.80	89.01±0.99	92.17±0.98	94.19±0.99	87.80±0.97	90.18±0.65	89.05±1.33	90.85±0.59	88.39±1.07	**95.11±0.14**

TABLE VIII

CLASSIFICATION RESULTS (%) FOR THE HYRANK DATASET (TEN PERCENT LABELED SAMPLE PER CLASS)

Class No.	Supervised methods or semi-supervised methods							Unsupervised or self-supervised methods					
	SVM [1]	3D-CNN [10]	BASSNet [53]	MSDCNet	ALPN [54]	Semi-LG-AGCN [38]	DSCA-Net [55]	VAE [43]	SimSiam [30]	DMVL [26]	ContrastiveNet [25]	SC-EADNet [50]	SS-MSDCNet (SVM)
1	63.41±3.37	65.27±3.76	65.64±11.14	66.18±6.48	84.34±6.75	74.35±9.79	72.25±6.26	79.27±3.96	77.71±7.43	73.97±4.35	65.17±4.75	74.85±6.42	**88.29±5.02**
2	77.86±3.61	87.16±8.29	87.33±8.92	99.00±3.00	**100.00±0.00**	98.73±2.44	**100.00±0.00**	97.33±4.67	98.20±3.17	95.69±4.36	71.67±12.63	99.84±0.49	**100.00±0.00**
3	67.39±1.36	87.58±4.26	73.85±6.91	71.29±4.72	88.21±4.75	**91.52±3.51**	88.63±4.17	85.94±2.95	82.90±3.32	83.28±2.41	67.79±3.51	81.48±4.87	88.87±2.93
4	50.13±9.64	53.48±8.35	54.08±7.21	31.41±11.43	**90.08±9.24**	41.22±12.21	51.55±15.92	66.76±11.13	65.33±10.11	49.25±10.20	48.59±12.87	73.92±10.07	83.38±9.17
5	61.73±0.95	90.18±0.95	82.70±4.71	87.30±3.04	85.94±3.23	**93.44±1.38**	93.11±4.10	81.51±2.69	84.47±2.12	81.11±1.44	86.60±2.18	78.13±4.69	88.97±1.57
6	44.08±5.81	55.94±6.25	45.67±9.90	36.87±10.18	69.24±6.99	50.24±9.32	51.86±8.13	70.70±6.81	72.08±8.38	62.74±8.56	45.82±6.37	65.60±9.67	**83.43±5.72**
7	52.38±2.32	63.58±4.61	61.31±6.37	40.47±4.75	76.28±3.06	61.63±6.39	66.92±3.31	69.31±3.52	**86.00±5.40**	80.02±4.25	65.07±4.55	79.35±2.29	85.68±3.36
8	53.03±7.39	64.83±5.79	53.88±9.88	86.41±4.34	71.63±5.51	74.61±8.95	84.05±2.89	83.50±4.07	84.87±2.41	89.00±1.41	83.32±2.83	75.36±4.03	**89.15±2.56**
9	72.57±2.53	84.58±2.17	83.70±3.65	83.70±3.63	67.85±2.62	86.51±2.98	87.20±1.20	64.00±2.76	81.47±1.53	80.51±1.68	**93.66±1.11**	75.93±3.23	81.49±1.16
10	74.21±1.57	83.66±1.83	79.41±6.16	82.65±3.98	64.01±4.93	83.02±3.05	85.65±2.02	70.46±2.33	84.58±1.65	83.35±1.07	**87.46±1.65**	77.04±2.46	87.28±1.64
11	59.48±6.98	73.52±5.31	53.11±9.41	80.80±3.50	85.64±3.86	79.68±8.20	85.34±7.68	94.16±4.92	91.85±6.63	90.91±3.94	83.00±5.21	80.88±5.34	**95.47±2.85**
12	85.94±1.65	90.19±2.04	91.74±2.69	89.18±3.31	93.38±2.51	95.32±1.73	90.76±5.80	95.37±2.86	90.00±2.78	93.12±3.40	86.46±3.98	95.27±2.73	**97.32±1.24**
13	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.88±0.34	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.69±0.40	**100.00±0.00**	99.28±0.75	99.86±0.26	**100.00±0.00**
14	99.86±0.04	99.57±0.25	99.98±0.07	99.98±0.07	99.68±0.37	**100.00±0.00**	**100.00±0.00**	99.53±0.73	95.19±3.63	97.57±1.35	93.87±2.18	99.41±0.94	**100.00±0.00**
OA(%)	69.71±0.43	83.59±0.43	78.52±0.73	82.93±0.57	76.35±0.86	85.73±1.11	87.33±0.88	77.33±0.56	85.51±0.64	84.73±0.58	86.96±0.39	79.79±0.79	**87.96±0.29**
AA(%)	67.82±0.61	81.85±0.79	76.15±1.26	75.37±1.49	84.01±1.53	80.73±1.80	82.67±1.44	82.70±0.88	85.31±2.18	82.90±1.64	76.98±1.07	82.64±2.85	**89.63±1.47**
kappa(%)	66.83±0.59	83.74±0.49	74.20±0.91	79.63±0.74	71.99±1.02	82.98±1.31	84.89±1.04	73.61±0.64	82.74±0.76	81.79±0.70	84.22±0.48	75.81±1.02	**85.61±0.35**

extracting robust spectral–spatial features. However, by equipping with the designed self-supervised training, SS-MSDCNet achieves better classification performance on all three datasets. In particular, SS-MSDCNet outperforms the state-of-the-art semisupervised learning method DSCA-Net. Therefore, it is evident that some advanced unsupervised feature learning methods can achieve performance levels that match or even surpass those of classical supervised or semisupervised learning methods. Furthermore, we observe from the experiments that on datasets such as Hyrank-Loukia, which have imbalanced class distributions and scattered class samples, SS-MSDCNet exhibits superior classification performance in some classes with smaller sample sizes, such as class Nos. 4 and 6 in the Hyrank-Loukia dataset.

Compared to the unsupervised or self-supervised methods, SS-MSDCNet still demonstrates superior classification

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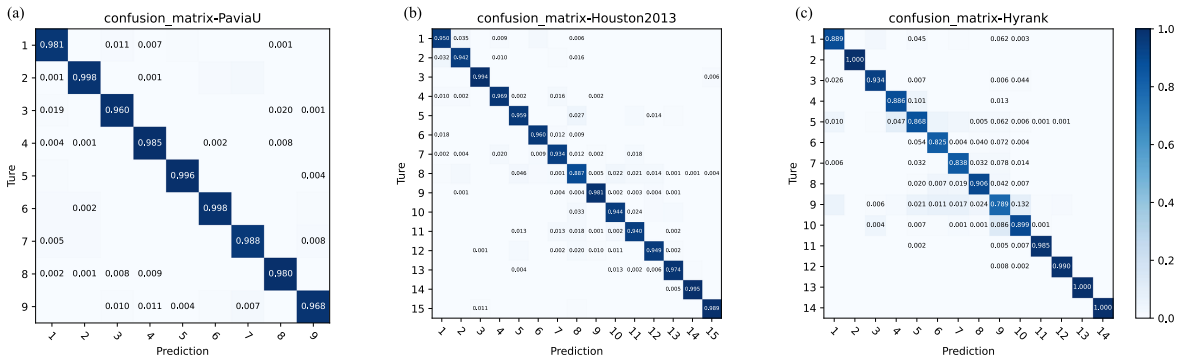


Fig. 9. Confusion matrices. (a) PaviaU. (b) Houston2013. (c) Hyrank.

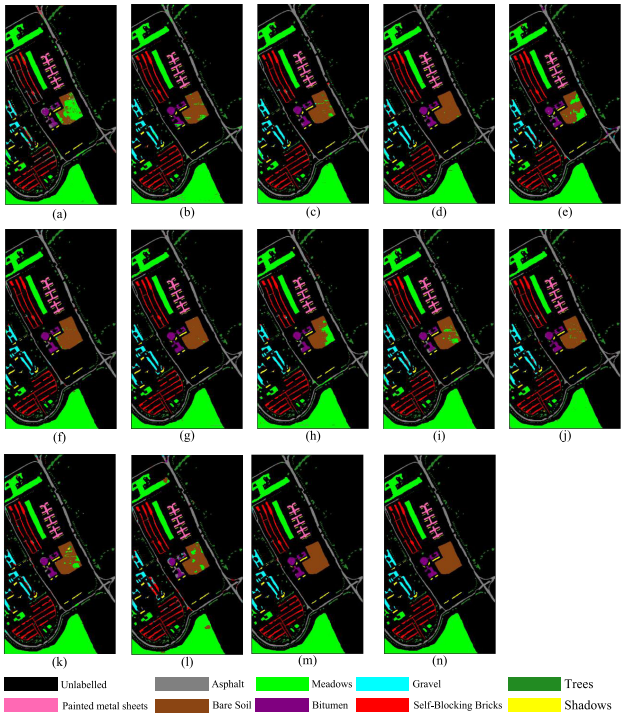


Fig. 10. Classification maps obtained via various methods on the Pavia University dataset. (a) SVM. (b) 3D-CNN. (c) BASSNet. (d) MSDCNet. (e) ALPN. (f) Semi-LG-AGCN. (g) DSCA-Net. (h) VAE. (i) SimSiam. (j) DMVL. (k) ContrastiveNet. (l) SC-EADNet. (m) SS-MSDCNet (SVM). (n) Ground truth.

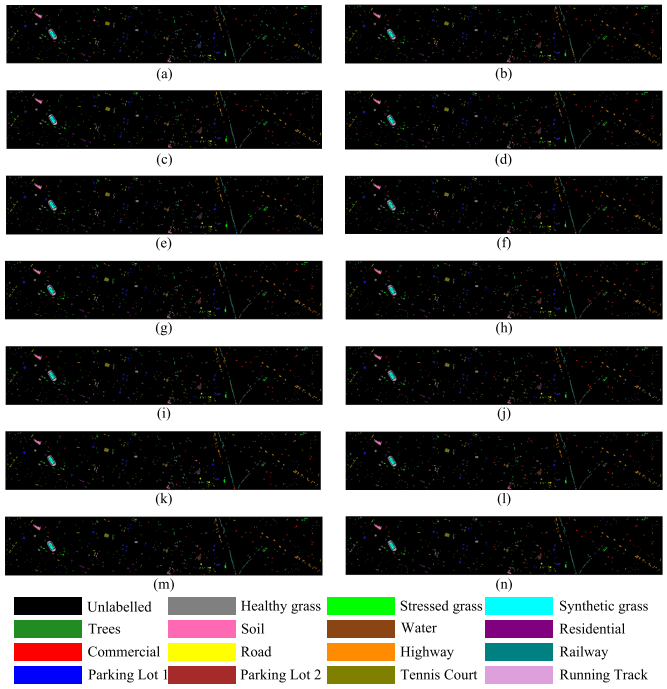


Fig. 11. Classification maps obtained via various methods on the Houston2013 dataset. (a) SVM. (b) 3D-CNN. (c) BASSNet. (d) MSDCNet. (e) ALPN. (f) Semi-LG-AGCN. (g) DSCA-Net. (h) VAE. (i) SimSiam. (j) DMVL. (k) ContrastiveNet. (l) SC-EADNet. (m) SS-MSDCNet (SVM). (n) Ground truth.

performance. Compared to generative models such as VAE, SS-MSDCNet achieves better classification results. For example, SS-MSDCNet outperforms VAE by 10.63% in terms of OA on the Hyrank-Loukia dataset. Compared to the contrastive self-supervised counterparts, i.e., DMVL, ContrastiveNet, and SC-EADNet, which typically rely on the use of positive and negative sample pairs, SS-MSDCNet does not require negative sample pairs. In Tables VI–VIII, SS-MSDCNet shows better performance on all three datasets than DMVL, ContrastiveNet, and SC-EADNet, which confirms that SS-MSDCNet can avoid model collapse even without the need for negative samples and achieve better classification performance. In addition, compared to SimSiam, SS-MSDCNet still demonstrates improved classification accuracy on all three datasets. This indicates that our designed backbone network, MSDCNet, and data augmentation techniques tailored for HSIs are effective in learning robust features.

Fig. 9 displays the confusion matrices for SS-MSDCNet on all three datasets. It can be seen that the probability of misclassification for SS-MSDCNet in the Pavia University dataset and the Houston2013 dataset is very small. To be specific, for the Pavia University dataset, SS-MSDCNet achieves an accuracy close to 100% for the *Meadows*, *Painted metal sheets*, and *Bare Soil* classes. For the Houston2013 dataset, SS-MSDCNet achieves an accuracy close to 100% for the *Synthetic grass* and *Tennis Court* classes. The probability of misclassification of the Hyrank-Loukia dataset is high due to the large difference in the number of samples among different classes. However, the accuracy of SS-MSDCNet can achieve 100% in some classes, such as *Mineral Extraction Sites*, *Water 2*, and *Coastal Water* classes.

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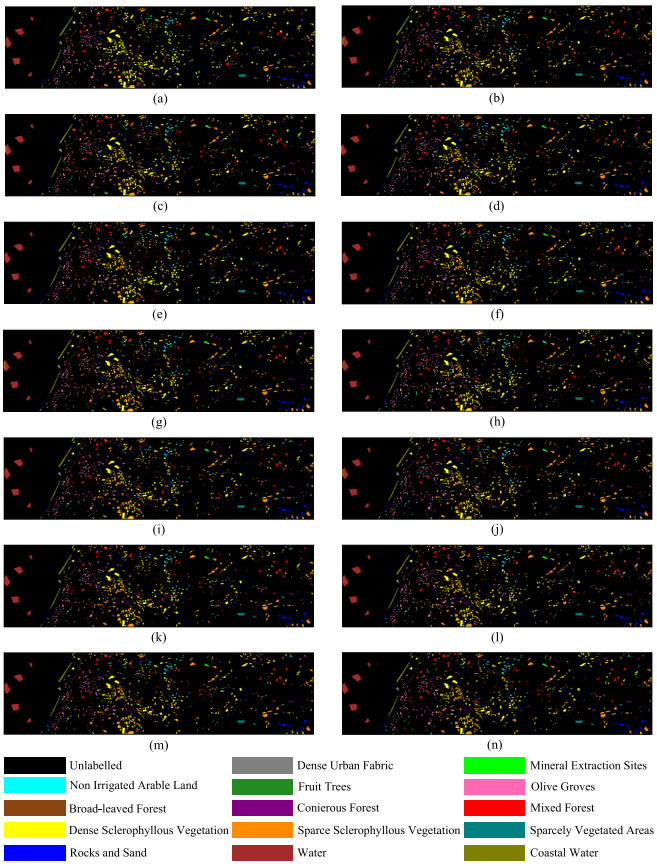


Fig. 12. Classification maps obtained via various methods on the Hyrank dataset. (a) SVM. (b) 3D-CNN. (c) BASSNet. (d) MSDCNet. (e) ALPN. (f) Semi-LG-AGCN. (g) DSCA-Net. (h) VAE. (i) SimSiam. (j) DMVL. (k) ContrastiveNet. (l) SC-EADNet. (m) SS-MSDCNet (SVM). (n) Ground truth.

For visual comparison, Figs. 10–12 show the classification maps of the various methods on the three datasets, where the maps produced by SS-MSDCNet match the ground-truth map best, further demonstrating the superiority of SS-MSDCNet. Compared to other methods, the classification maps generated by SS-MSDCNet are the smoothest with the least noise. Taking the Hyrank dataset as an example, it has lower spatial resolution and uneven class distribution compared to other datasets, which is hard to classify. However, compared to other methods, SS-MSDCNet has demonstrated impressive classification performance on this dataset.

D. Transfer Experiment

To study the generalization ability of SS-MSDCNet, we perform self-supervised training on a new dataset, the WHU-Hi-HongHu dataset. Subsequently, we transfer the pretrained model to three different datasets: Pavia University, Houston2013, and Hyrank-Loukia datasets for classification tasks. These datasets are obtained from distinct sensors with different spatial and spectral resolutions. The WHU-Hi-HongHu dataset boasts an impressive spatial resolution of 0.043 m, making it an ideal choice for self-supervised pretraining. The other three datasets have a maximum spatial resolution of only 1.3 m, found in the Pavia University dataset, and the Hyrank-Loukia dataset, which is acquired using satellite sensors, has an image spatial resolution of just 30 m.

The classification results on the three datasets by the proposed self-supervised training on their own datasets (i.e., no transfer) and on the WHU-Hi-HongHu datasets (transfer) and without the self-supervised training (i.e., supervised training) are listed in Table IX. Typically, when there is a substantial difference between the datasets used for self-supervised training and testing, it often results in suboptimal classification performance. However, as presented in Table IX, the proposed SS-MSDCNet still exhibits strong performance on a dataset by self-supervised training on another dataset despite their significantly different characteristics. For example, by employing self-supervised training, the classification results with and without transfer on the Pavia University and Hyrank datasets are nearly the same, and they all outperform the case without the self-supervised training. Although there is a slight drop in classification performance for the Houston2013 dataset possibly due to its huge difference from the WHU-Hi-HongHu dataset, the classification performance still matches that of advanced unsupervised classification methods, such as DMVL and ContrastiveNet, as listed in Tables VI–VIII. Therefore, SS-MSDCNet demonstrates excellent generalization performance. The backbone network of SS-MSDCNet effectively leverages the supervisory information acquired from self-supervised training on unlabeled data, and this knowledge remains effective when applied to other datasets.

E. Ablation Study

The backbone network for the proposed SS-MSDCNet is MSDCNet, which consists of a multibranch structure with a 3-D densely connected module and a 3-D spatial attention module. In this section, we conduct a series of ablation experiments to validate the significance of these designs within the MSDCNet.

First, we study the importance of the multibranch structure within MSDCNet by configuring networks with different numbers of branches. In Table X, the three-branch structure in MSDCNet yields significant improvements across all three datasets compared to the single-branch and double-branch configurations. On the Pavia University dataset, the three-branch structure exhibits an improvement of 2.86% over the double-branch configuration and an improvement of 7.51% over the single-branch configuration in terms of OA. In the Houston dataset, the three-branch structure shows an average OA improvement of 2.37% over the double-branch configuration and an improvement of 4.56% over the single-branch configuration. In the Hyrank-Loukia dataset, the three-branch structure demonstrates an average OA improvement of 2.78% points over the double-branch configuration and an improvement of 4.62% over the single-branch configuration. From this, it is evident that the three-branch structure utilized in this article is better at leveraging pixel-level contextual information from HSIs compared to the double- and single-branch configurations. Furthermore, we design an ablation experiment to validate the importance of two main blocks used in the MSDCNet.

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TABLE IX
EXPERIMENTAL RESULTS OF THE TRANSFER EXPERIMENT

Datasets	Pavia University			Houston2013			Hyrank		
	Unsupervised		Supervised	Unsupervised		Supervised	Unsupervised		Supervised
	No transfer	Transfer		No transfer	Transfer		No transfer	Transfer	
OA(%)	98.87±0.16	98.04±0.22	97.00±1.44	95.47±0.13	93.55±0.62	94.06±0.74	87.96±0.29	86.64±0.89	82.93±0.57
AA(%)	98.43±0.33	97.58±0.46	95.73±1.53	95.87±0.12	94.18±0.69	93.94±0.65	89.63±1.47	86.55±1.91	75.37±1.49
Kappa(%)	98.51±0.21	97.40±0.31	95.98±1.97	95.11±0.14	93.03±0.67	93.58±0.80	85.61±0.35	84.09±1.06	79.63±0.74

TABLE X
RESULTS OF THE MSDC BRANCHING STRUCTURE ABLATION STUDY

	Single branch	Dual branchs	Three branchs
Pavia University	91.36±0.42	96.01±0.22	98.87±0.16
Houston2013	90.91±0.64	93.10±0.49	95.47±0.13
Hyrank-Loukia	83.34±0.84	85.18±0.68	87.96±0.29

TABLE XI
RESULTS OF MODULES' ABLATION STUDY

	3DDC	3DSA	PaviaU	Houston2013	Hyrank-Loukia
Network A			93.76±0.28	90.89±0.53	85.08±0.54
Network B	✓		94.26±0.29	91.33±0.65	86.48±0.99
Network C		✓	94.79±0.35	92.44±0.47	86.54±0.71
SS-MSDCNet	✓	✓	98.87±0.16	95.47±0.13	87.96±0.29

As shown in Table XI, three different networks were designed to measure the importance of the two main blocks in the MSDCNet. For Network A, a module based on a 3-D Resnet block is employed to construct the network. For Network B, only a 3-D densely connected module is employed to construct the network. For Network C, a 3-D spatial attention module is added on the basis of network A. In Table XI, compared with Network A, the OAs of Network B have improved for three datasets. This indicates that the 3-D densely connected module has stronger feature extraction capabilities compared to the regular residual connection module. Similarly, Network C also outstands Network A. This suggests that the 3-D spatial attention module assists the network in focusing on information more conducive to classification. In comparison to the three designed networks, SS-MSDCNet demonstrates optimal classification performance for all three datasets. Therefore, the use of the 3-D densely connected module and the 3-D spatial attention module in MSDCNet enhances the network's ability to learn information from HSIs.

During the self-supervised training phase of the proposed SS-MSDCNet, we design a data augmentation method based on spectral splitting to coordinate the two-stream structure of SS-MSDCNet, aiming to fully utilize the abundant spectral information of HSIs. In this section, we also conduct an ablation study to validate the effectiveness of the designed data augmentation method. According to the experimental results shown in Fig. 13, it is evident that after incorporating the designed data augmentation method, SS-MSDCNet exhibits

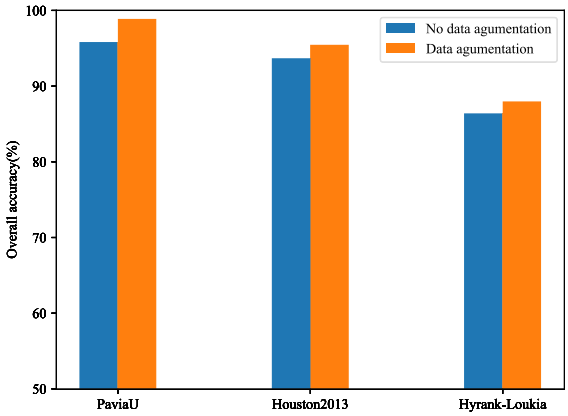


Fig. 13. Results of data augmentation ablation study.

TABLE XII
PARAMETER AND FLOPS UNDER DIFFERENT METHODS

	SimSiam	DMVL	SC-EADNet	SS-MSDCNet
Parameter(M)	24.62	24.62	1.51	8.02
FLOPs(M)	545.82	1075.2	10.51	1351.68

significant improvements in classification performance for three datasets. Therefore, the data augmentation method based on spectral splitting effectively utilizes the abundant spectral information in HSIs, which is conducive to the HSI classification task.

F. Parameter Analysis

The complexity of the deep learning model structure can be measured by parameter numbers and floating point operations (FLOPs) of the network. Table XII lists the parameter numbers and FLOPs of SS-MSDCNet and several representative self-supervised learning methods. It can be found that the proposed SS-MSDCNet has lower parameter numbers compared with the two self-supervised learning methods, SimSiam and DMVL, which use ResNet51 as the backbone. It can also be found that SS-MSDCNet has higher FLOPs, which is due to the three-branch structure of MSDCNet and the two-stream structure of the self-supervised framework. In addition, SC-EADNet has the lowest parameter numbers and FLOPs mainly because it adopts a lightweight and efficient backbone. However, the backbone of SC-EADNet does not fully utilize the spatial–spectral features of HSI, which leads to the worst classification performance among the four methods.

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The experimental results in Tables VI–VIII show that the proposed SS-MSCNet has a large improvement in classification performance compared with the other three self-supervised methods in Table XII. Therefore, it can be concluded that SS-MSCNet achieves a balance between model complexity and classification performance.

V. Conclusion

This article introduces a novel self-supervised HSI classification method, named SS-MSDCNet. This method allows for the unsupervised learning of useful representations from unlabeled samples of HSIs. In SS-MSDCNet, a two-stream self-supervised framework is designed, accompanied by a data augmentation technique based on spectral splitting. This allows SS-MSDCNet to extract spectral information from different bands of HSIs. In addition, a backbone network is designed, named MSDCNet, which not only effectively leverages pixel-level contextual information from HSIs but also takes full advantage of the multiscale features extracted by various convolutional layers. Furthermore, MSDCNet utilizes a 3-D spatial attention module to assign weights to the spatial features of each pixel, allowing the network to better learn information. We have substantiated the excellent classification and generalization performance of SS-MSDCNet through comparative and transfer experiments. What is more, we conducted ablation experiments to verify the effects of the data augmentation technique and the main modules of the backbone network. In the future, we plan to design more efficient and lightweight networks for HSI self-supervised learning.

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